



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc DEGREE EXAMINATION
COLLEGE: SCIENCE AND TECHNOLOGY
SCHOOL: NATURAL AND APPLIED SCIENCES
DEPARTMENT: COMPUTER AND INFORMATION SCIENCES
SESSION: 2015/2016

SEMESTER: ALPHA

COURSE CODE: CSC 213

CREDIT

UNIT: 3

COURSE TITLE: STRUCTURED PROGRAMMING

INSTRUCTION: ANSWER ANY FOUR QUESTIONS

Time: 2

$\frac{1}{2}$ Hours

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1. a) What is Structured Programming? **1.5 marks**
- b) A program is Structured if it uses _____ and _____. **2 marks**
- c) Give Three (3) differences between Structured and Object Oriented Programming. **3 marks**
- d) What is an Algorithm. Mention the fundamental properties of a good Algorithm. **4 marks**
- f) Write the algorithm and draw the flow chart to compute the GPA for a student using the formula $GPA = \frac{\text{Total_GP}}{\text{Total_Credit}}$ **7 marks**
2. a) Given;
- ```
>>> a = set ('abracadabra')
>>> b = set ('alacazam')
```
- What is
- `>>> a - b`
  - `>>> a | b`
  - `a & b`
  - `>>> a ^ b`
- 2Marks**
- b) Give the syntax and examples of the following statements in python;
- If statement
  - The if statement with alternative execution
  - Chained conditional statements
  - While statements
- 6Marks**
- c) Write a program that converts Fahrenheit to centigrade using the formula  $c = (5/9) * (F - 32)$  and stores the result in a file like this

| F    | C              |
|------|----------------|
| 41.0 | 4.444444444444 |
| 42.0 | 5.0            |
| 43.0 | 5.555555555556 |
| 44.0 | 6.111111111111 |
| 45.0 | 6.666666666667 |
| 46.0 | 7.222222222222 |
| 47.0 | 7.777777777778 |
| 48.0 | 8.333333333333 |
| 49.0 | 8.888888888889 |
| 50.0 | 9.444444444444 |

**7Marks**

d) Write the python program to display the following interface;



**2<sup>1</sup>/<sub>2</sub>Marks**

3. a) Explain the following and give examples

- i. Tuples
- ii. Dictionary
- iii. Sets

**4<sup>1</sup>/<sub>2</sub>Marks**

b) Write a program in python to check if a number is even. (input 2 output true, input 3 : output false). Make sure your program takes care of invalid inputs

**4Marks**

c) Implement the linear search algorithm in python such that you can search for members of the following array, declared and initialized in your program 13,0,1,7,4

**9Marks**

4. a) Given;

```
tuple= ('abcd', 786, 2.23, 'john', 70.2)
tinytuple =(123 , 'john')
```

What is the output of;

- i. print tuple[0]
- ii. print tuple[1:3]
- iii. print tuple[2:]
- iv. print tuple\*2
- v. print tuple + tinytuple

**2<sup>1</sup>/<sub>2</sub>Marks**

b) Write a python program to calculate the sum of the first 10 terms of the series.

1/1 + 1/4 + 1/9 + 1/16.....

**4Marks**

c) Write a program that generates a table of factorials FROM 0 TO 10 like this

```
0!=0
1!=1
2!=2
3!=6
4!=24
```

**5Marks**

d) The following are used for what in Tkinter;

- i. The **pack** Method

- ii. The *grid* Method
- iii. The *place* Method.

**6 Marks**

5. a) Write a python program to design a GUI like this



- i. To receive two numbers
- ii. Add, subtract, multiply or divide them
- iii. Displays the result of any action taken by clicking any of the buttons.

**10 Marks**

- b) Explain in details the concept of Modularity in structured programming. **3<sup>1</sup>/<sub>2</sub>Marks** c)

Give a detailed overview of python programming State four application domains with examples of Python programming language.

**4 Marks**

6. a) Create a file object in Python, and write “Covenant University is Great” to a file named **output.txt**

**1.5Marks**

- b) What is the output of the program below?

```
data=[]
row1=[]
row2=[]
average=[["row1"],["row2"]]

for i in range(0,3):
 row1.append(math.pow(i,2))
 row2.append((math.pow(i,2)+i))

data.append(row1)
data.append(row2)

for i in range (0,len(data)):
 add=0
 avg=0
 for j in range (0, len(data[0])):
 add+=data[i][j]
 avg=add/len(data[0])
 average[i].append(avg)

print (data)
print (average)
```

**4 Marks**

- c) Given the List Below:

```
myList=[["Orange", "Apple", "Banana"],
 ["Vitamin C", "Vitamin A", "Potassium"],
 [["5kg", "2kg"], ["4kg", "1kg", "3kg"], ["3kg", "2kg", "9kg"]]]
```

What would be the outputs of the following **print** statements?

**8 Marks**

- i. **print (myList)**
  - ii. **print (len(myList))**
  - iii. **print (len(myList[2]))**
  - iv. **print (len(myList[2][0]))**
  - v. **print (myList[1])**
  - vi. **print (myList[2])**
  - vii. **print (myList[1][1])**
  - viii. **print (myList[2][0]+ myList[2][1]+myList[2][2])**
- d) Convert the following mathematical expression to a function called 'EUD' in Python.

$$\sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

**4 Marks**